Estimation and Filtering

MCMC Estimation of Friction Parameters for Robotic Arms

Baldini Filippo Berti Federico Michielon Samuele

Ferracane Salvatore Vianello Giacomo

1 Introduction

This report aims at employing the Markov Chain Monte Carlo (MCMC) strategy in order to perform the estimation of the joint friction values of a robotic arm consisting of 2 revolute joints (2R manipulator).

Firstly, the dynamical model of the physical system under analysis is presented, showing how the unknown parameters are featured in it. Then, MCMC estimation process is implemented in MATLAB in order to obtain the results. These latter are discussed and the efficiency of the proposed method is evaluated.

2 Model of the Robotic Arm

We consider a planar 2-degree-of-freedom (2-DoF) robotic arm with two revolute joints, described by the joint angle vector $\boldsymbol{q} = [\theta_1 \ \theta_2]^\top$, as illustrated in Figure 1. Its dynamics follow the standard form

$$\boldsymbol{B}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{\tau}_{\text{friction}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau},$$

where $\boldsymbol{B}(\boldsymbol{q})$ is the inertia matrix, $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ accounts for Coriolis and centrifugal effects, $\boldsymbol{g}(\boldsymbol{q})$ is the gravity vector, and $\boldsymbol{\tau}$ represents the applied joint torques.

The friction torque vector $\boldsymbol{\tau}_{\text{friction}}$ combines viscous and Coulomb friction components, classically modeled as

$$\boldsymbol{\tau}_{\text{friction}} = \boldsymbol{b} \cdot \dot{\boldsymbol{q}} + \boldsymbol{\tau}_c \cdot \text{sign}(\dot{\boldsymbol{q}}),$$

where $\boldsymbol{b}$ are viscous friction coefficients and $\boldsymbol{\tau}_c$ are Coulomb friction levels. To avoid numerical issues due to the discontinuous sign$(\cdot)$ function, we adopt a smooth approximation

$$\boldsymbol{\tau}_{\text{friction}} = \boldsymbol{b} \cdot \dot{\boldsymbol{q}} + \boldsymbol{\tau}_c \cdot \tanh(\boldsymbol{k} \cdot \dot{\boldsymbol{q}}),$$

where $\boldsymbol{k}$ controls the sharpness of the Coulomb friction transition. Assuming perfect knowledge of $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{g}$, noisy torque measurements $\boldsymbol{\tau}'$ are modeled as

$$\boldsymbol{\tau}' = \boldsymbol{\tau} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}),$$

with known noise covariance $\boldsymbol{\Sigma}$. Hence, the friction torque model for estimation becomes

$$\boldsymbol{\tau}' = \boldsymbol{b} \cdot \dot{\boldsymbol{q}} + \boldsymbol{\tau}_c \cdot \tanh(\boldsymbol{k} \cdot \dot{\boldsymbol{q}}) + \boldsymbol{\epsilon}. \tag{1}$$

The aim of the subsequent estimation process is to identify the friction parameters $\boldsymbol{b}$, $\boldsymbol{\tau}_c$, and $\boldsymbol{k}$ from measured joint velocities and noisy torque observations.

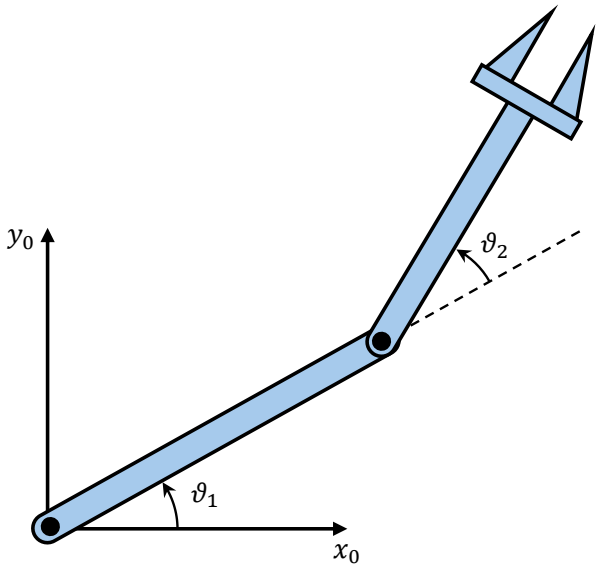


Figure 1: Sketch of the 2R Robotic Arm.

3 Estimation Procedure

This section presents a Bayesian framework for the identification of joint friction parameters in the robotic system described earlier. The estimation is performed using the Metropolis-Hastings algorithm, a widely used Markov Chain Monte Carlo (MCMC) technique capable of sampling from complex, high-dimensional, and non-Gaussian posterior distributions.

As explained previously, the dynamic model of the manipulator is assumed to be known exactlys so that the only unknown components in the joint dynamics are the friction torques. This assumption simplifies the estimation process, as the known dynamic terms can be subtracted from the measured joint torques. Hence, the residual torques correspond exclusively to the effects of joint friction.

The objective is to estimate the six friction parameters

$$\boldsymbol{\theta} = \begin{bmatrix} b_1 & b_2 & \tau_{c1} & \tau_{c2} & k_1 & k_2 \end{bmatrix}^\top \in \mathbb{R}^6,$$

from noisy measurements of joint velocities and actuation torques. The Bayesian formulation provides not only point estimates (e.g., posterior means), but also quantifies uncertainty via the full posterior distributions of the parameters.

To ensure physical plausibility (i.e., parameter positivity) and to enhance numerical stability, sampling is performed in logarithmic space. That is, the MCMC chain evolves over transformed variables

$$\boldsymbol{\phi} = \log \boldsymbol{\theta},$$

and is mapped back to the original parameter domain using the exponential function. This transformation allows for unconstrained sampling in $\mathbb{R}^6$, while implicitly enforcing positivity of the parameters.

MCMC Estimation from Simulated Data

To validate the proposed estimation strategy, we perform a simulation-based study using synthetic data. The estimation procedure is structured as follows:

1. Simulate joint trajectories and torque signals using known (ground truth) values of the friction parameters.

2. Add zero-mean Gaussian noise with known covariance to the torque data to emulate sensor imperfections.

3. Initialize the MCMC chain in log-space with a reasonable initial guess, and run the Metropolis-Hastings algorithm for 100,000 iterations.

4. Discard the initial 10% of the chain as burn-in, then analyze the remaining samples to compute posterior statistics and evaluate estimation accuracy.

The next subsection introduces the construction of the log-likelihood function, which forms the core of the posterior evaluation in the MCMC sampling process.

Log-Likelihood Function

The `log_likelihood` function computes the unnormalized log-posterior probability for a given candidate vector of friction parameters. It quantifies how well the proposed parameters explain the observed data under the assumed noise and model structure.

Let the parameter vector in log-space be

$$\boldsymbol{\phi} = \begin{bmatrix} \log b_1 & \log b_2 & \log \tau_{c1} & \log \tau_{c2} & \log k_1 & \log k_2 \end{bmatrix}^\top \in \mathbb{R}^6, \tag{2}$$

and define the corresponding physical parameters via

$$\boldsymbol{\theta} = \exp(\boldsymbol{\phi}) = \begin{bmatrix} b_1 & b_2 & \tau_{c1} & \tau_{c2} & k_1 & k_2 \end{bmatrix}^\top.$$

We denote the dataset of joint velocity and torque measurements as

$$\mathcal{D} = \left\{ \dot{\boldsymbol{q}}_i,\ \boldsymbol{\tau}_i^{\text{obs}} \right\}_{i=1}^{N}, \tag{3}$$

where $\dot{\boldsymbol{q}}_i = [\dot{q}_{i1}, \dot{q}_{i2}]^\top$ are the joint velocities and $\boldsymbol{\tau}_i^{\text{obs}} = [\tau_{i1}^{\text{obs}}, \tau_{i2}^{\text{obs}}]^\top$ the observed torques. For each sample i, the model-predicted torque based on the nonlinear friction model is given by

$$\boldsymbol{\tau}_i^{\text{model}}(\boldsymbol{\theta}) = \begin{bmatrix} b_1 \dot{q}_{i1} + \tau_{c1} \tanh(k_1 \dot{q}_{i1}) \\ b_2 \dot{q}_{i2} + \tau_{c2} \tanh(k_2 \dot{q}_{i2}) \end{bmatrix}. \tag{4}$$

Assuming additive, zero-mean Gaussian noise with known covariance matrix

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{2\times 2},$$

the residuals are modeled as

$$\boldsymbol{\epsilon}_i = \boldsymbol{\tau}_i^{\text{obs}} - \boldsymbol{\tau}_i^{\text{model}}(\boldsymbol{\theta}), \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \Sigma).$$

The likelihood function over the full dataset is then

$$p(\mathcal{D} \mid \boldsymbol{\theta}) = \prod_{i=1}^{N} \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} \boldsymbol{\epsilon}_i^\top \Sigma^{-1} \boldsymbol{\epsilon}_i \right), \tag{5}$$

where $d = 2$ is the dimension of the torque vector. Taking logarithms yields the log-likelihood

$$\log p(\mathcal{D} \mid \boldsymbol{\theta}) = -\frac{N}{2} \log \left((2\pi)^d |\Sigma| \right) - \frac{1}{2} \sum_{i=1}^{N} \boldsymbol{\epsilon}_i^\top \Sigma^{-1} \boldsymbol{\epsilon}_i. \tag{6}$$

To incorporate prior knowledge, we define a multivariate Gaussian prior on the log-parameters

$$\log p(\boldsymbol{\phi}) = -\frac{1}{2} (\boldsymbol{\phi} - \boldsymbol{\mu}_{\text{prior}})^\top \Sigma_{\text{prior}}^{-1} (\boldsymbol{\phi} - \boldsymbol{\mu}_{\text{prior}}). \tag{7}$$

Therefore, summing (6) to (7), the resulting unnormalized log-posterior is

$$\log p(\boldsymbol{\phi} \mid \mathcal{D}) \propto \log p(\mathcal{D} \mid \boldsymbol{\theta} = \exp(\boldsymbol{\phi})) + \log p(\boldsymbol{\phi}). \tag{8}$$

This log-posterior is evaluated at each iteration of the Metropolis-Hastings algorithm to determine whether to accept or reject proposed parameter samples.

Metropolis-Hastings Sampling Procedure

Bayesian inference is carried out using the Metropolis-Hastings algorithm, implemented in the `mcmc_tanh_logspace.m` function. This method generates samples from the posterior distribution of the transformed parameters $\boldsymbol{\phi} \in \mathbb{R}^6$.

The sampling procedure follows these steps:

- **Initialization:** Start with an initial guess $\boldsymbol{\phi}^{(0)}$ in log-space.
- **Proposal:** At each iteration t, generate a candidate sample

  $$\boldsymbol{\phi}' = \boldsymbol{\phi}^{(t)} + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\text{prop}}),$$

  where Σ_{prop} is the proposal covariance matrix that in our case was set to

  $$\Sigma_{\text{prop}} = \text{diag}\left([0.00003,\ 0.00003,\ 0.0008,\ 0.0008,\ 0.004,\ 0.004]\right)$$

  Recall that its diagonal values are meant to be in the log-space.
- **Acceptance Ratio:** Compute the acceptance probability

  $$\alpha = \min \left(1, \exp \left(\log p(\boldsymbol{\phi}' \mid \mathcal{D}) - \log p(\boldsymbol{\phi}^{(t)} \mid \mathcal{D}) \right) \right).$$

- **Acceptance Step:** Accept $\boldsymbol{\phi}'$ with probability α, otherwise retain $\boldsymbol{\phi}^{(t)}$.

The Markov chain is run for 100,000 iterations, and the first 10% of samples are discarded as burn-in. The remaining samples are used to compute posterior summaries, such as means, standard deviations, and credible intervals.

Marginal histograms and trace plots are used to visualize the posterior distributions and assess convergence. Additionally, the posterior mean of each parameter is compared to the ground-truth values to evaluate estimation accuracy.

The log-space formulation ensures that the sampled physical parameters $\boldsymbol{\theta} = \exp(\boldsymbol{\phi})$ are always positive, removing the need for explicit constraints and improving numerical stability. This makes the method particularly suited for identifying nonlinear friction parameters in joint-level dynamics.

4 Results Discussion

Following the methodology described in the previous section, this chapter illustrates the results obtained by running the estimation procedure and evaluating the inferred parameter distributions.

First, it is important to determine whether the simulations parameters, such as the number of iteration and the burn-in lenght, were sufficient to ensure the validity of the results. This assessment is not straightforward and is often difficult to make prior to running the simulations, due to the complexity of the MCMC kernel. For the results to be considered valid, the algorithm must demonstrate good convergence.

The plots in Figure 2 illustrate the evolution of the algorithm iteration by iteration. They show a strong evidence of convergence, indicating that the algorithm parameters provided a reliable simulation process.

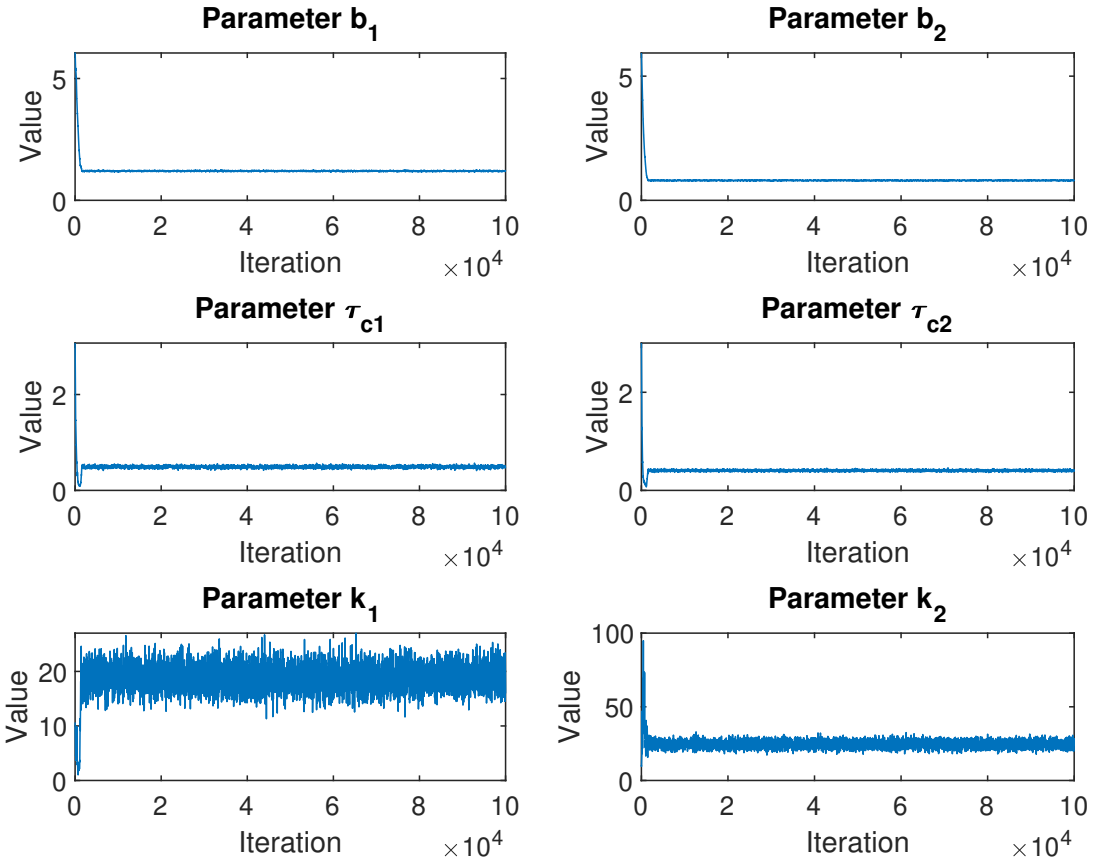


Figure 2: Evolution of the algorithm over iterations

The posterior distributions shown in Figure 4 exhibit well-defined, symmetric shapes centered around the true parameter values (indicated by red lines), confirming the reliability of the estimates. The smoothness and alignment of the samples with the ground truth suggest effective convergence and thorough exploration of the parameter space, highlighting the robustness of the MCMC sampling procedure.

Figure 3 displays the 95 % confidence intervals of the estimated parameters. As can be observed, all the linear parameters are estimated with high precision, while the nonlinear parameters, k_1 and k_2, exhibit wider confidence intervals—though they still include the true underlying values. This increased uncertainty is primarily attributed to the strong dependency of these parameters on the joint trajectories followed by the manipulator

during the dataset generation phase. Nonetheless, their final estimates remain close to the ground-truth values, indicating a good level of accuracy. A comprehensive summary of the final parameter estimates is provided in Table 1.

It is important to note that the dataset used for parameter estimation is generated afresh with each run of `main.m`. As a result, the estimation outcomes may vary slightly from run to run, although such variations are generally not significant.

In the considered case, the sampling procedure achieved an acceptance rate of

$$\text{Acceptance rate} = 26.62\,\%.$$

This value falls within the commonly recommended range for Metropolis-Hastings algorithms (approximately 20%–40%), indicating a well-balanced trade-off between exploration of the parameter space and sampling efficiency. An acceptance rate in this range suggests that the proposed candidates are sufficiently diverse to ensure good exploration, while being accepted frequently enough to promote convergence and mixing of the Markov chain.

Parameter	**Estimate**	**True Value**	**Std. Dev.**	**Confidence Interval**
b_1	1.2029	1.200	0.0133	$[1.1766,\ 1.2294]$
b_2	0.7999	0.800	0.0082	$[0.7839,\ 0.8162]$
τ_1	0.4909	0.500	0.0191	$[0.4541,\ 0.5288]$
τ_2	0.4054	0.400	0.0118	$[0.3826,\ 0.4283]$
k_1	18.7654	20.000	1.9450	$[15.0328,\ 22.6335]$
k_2	24.4952	25.000	1.9738	$[20.6403,\ 28.3959]$

Table 1: Estimation results from Matlab simulation.

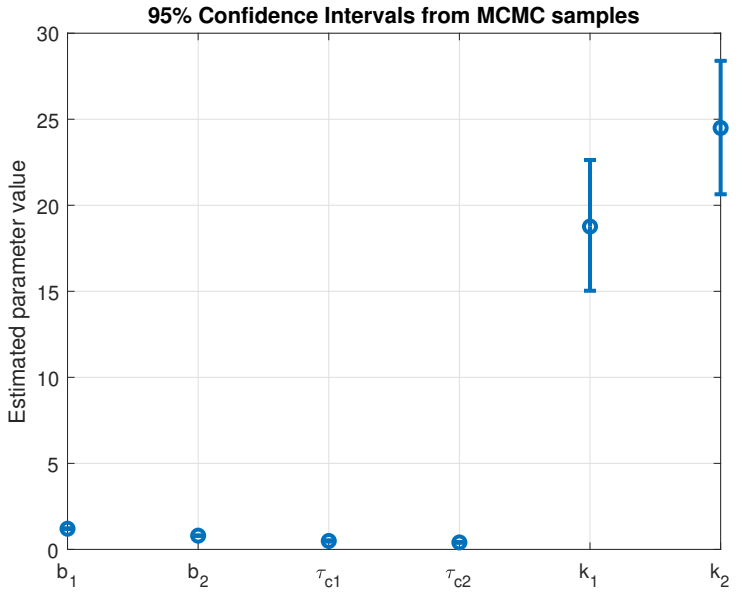
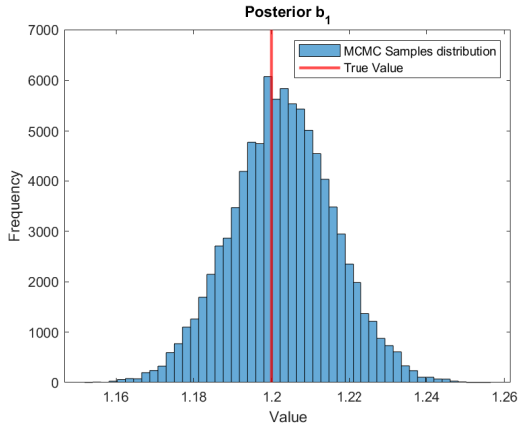
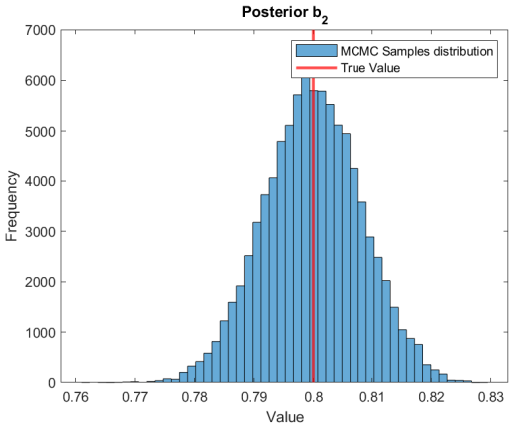


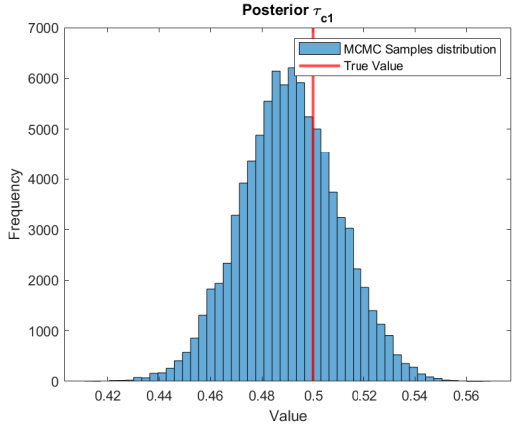
Figure 3: 95% confidence intervals of the proposed estimates.



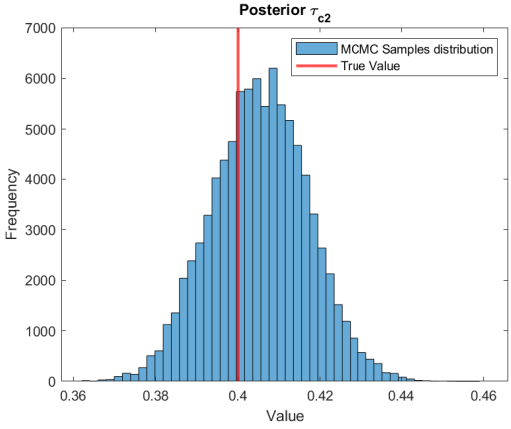
(a) b_1



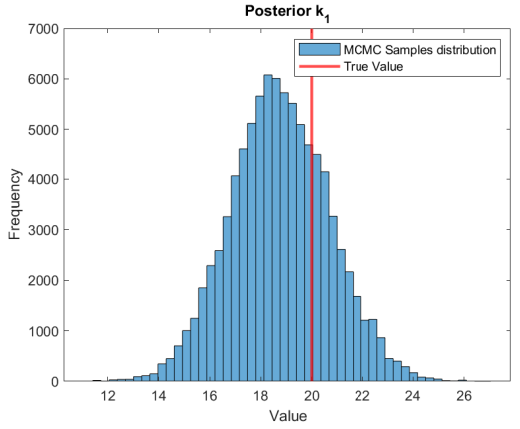
(b) b_2



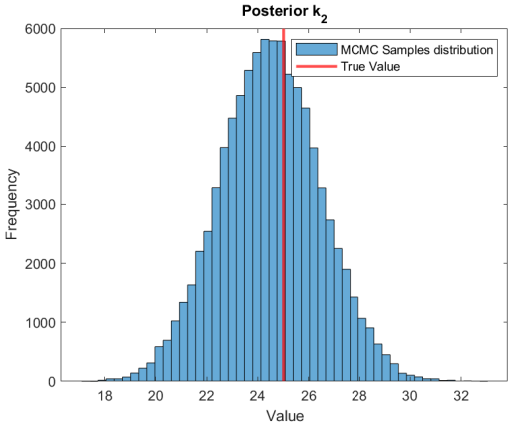
(c) τ_{c1}



(d) τ_{c2}



(e) k_1



(f) k_2

Figure 4: Parameter estimation results for friction parameters b_1, b_2, τ_{c1}, τ_{c2}, k_1, and k_2.